

Voltage Regulation

External Flash

The schematic shows an external flash memory circuit. A red rectangular component labeled U8 is connected to various signals. On the left, four signals are connected to pins 1 through 4: QSPI_SS to pin 1, QSPI_SD1 to pin 2, QSPI_SD2 to pin 3, and GND to pin 4. Pin 1 also has a pull-up resistor R4 (1K) connected to VCC. The bottom of the circuit is connected to GND. On the right, three signals are connected to pins 7 through 9: QSPI_SD3 to pin 7, QSPI_SCLK to pin 8, and QSPI_SD0 to pin 9. Pin 8 also has a pull-up resistor C12 (100nF) connected to VCC. The top of the circuit is connected to VCC.

Expansion Headers

The diagram illustrates the connection of two expansion headers, H2 and H3, to the PM254V-11-06-H85 module. Header H2 is connected to the module's pins 1 through 6 via the following signals: XIAOTX (pin 1), XIAOMOSI (pin 2), XIAOMISO (pin 3), XIAOSCL (pin 4), and XIAORX (pin 5). Header H3 is connected to the module's pins 1 through 6 via the following signals: XIAO3V3 (pin 1), SDA (pin 2), SCL (pin 3), and a GND connection (pin 4). The module is labeled PM254V-11-06-H85.

Crystal Oscillator

The diagram illustrates a crystal oscillator circuit. A 12MHz crystal (X2) is the central component. Its pin 1 is connected to the input signal XIN. Pin 3 is connected to the output signal XOUT. Pin 2 is connected to ground through a 1kΩ resistor (R5). Pins 4 and 5 are connected to ground through 15pF capacitors (C18 and C17 respectively).

Signal Multiplexers

Boot & Reset

The diagram illustrates the hardware connections for the Boot and Reset functions. It features two push buttons, SW1 and SW3, connected to a 3.3V supply. SW3 is connected to a 1K resistor (R3) and then to the QSPI_SS pin. SW1 is connected directly to the RUN pin. Both buttons have their other terminal connected to GND. The components are labeled SW1, SW3, R3 1K, QSPI_SS, and RUN. The ground symbol is labeled GND.

USB Header

The diagram illustrates the electrical connections for a USB header. A USB4105-GF-A-120 module is shown with its pinout on the right. The module's pins are connected to a USB header on the left. The header has pins for VUSB, GND, USB_D-, USB_D+, and a common ground. The module's pins are labeled as follows:

- Pin 1:** GND
- Pin 2:** VBUS
- Pin 3:** SBU2
- Pin 4:** CC1
- Pin 5:** Dn2
- Pin 6:** Dp1
- Pin 7:** Dn1
- Pin 8:** Dp2
- Pin 9:** SBU1
- Pin 10:** CC2
- Pin 11:** VBUS
- Pin 12:** GND

The connections are as follows:

- VUSB:** Connected to Pin 2 (VBUS) and Pin 11 (VBUS).
- GND:** Connected to Pin 1 (GND), Pin 12 (GND), and the common ground of the header.
- USB_D-:** Connected to Pin 7 (Dn1) and Pin 9 (SBU1).
- USB_D+:** Connected to Pin 6 (Dp1) and Pin 8 (Dp2).

Resistors R6 and R7 are shown in red boxes, connected between the header pins and the module pins. R6 is connected between the header pin for USB_D- and the module pin for Dn1. R7 is connected between the header pin for USB_D+ and the module pin for Dp1.

Telemetry Headers

The image displays three wiring diagrams for PM254-2-05-S-8.5 telemetry headers, labeled H4, H5, and H6. Each diagram shows a 10-pin header with pins numbered 1 through 10. Pins 1, 2, 3, 4, and 5 are on the left, and pins 10, 9, 8, 7, and 6 are on the right. The diagrams show the following connections:

- H4:** SW1WIRE (pin 1), DIGIT1 (pin 2), and LSWITCH (pin 3) are connected to pins 1, 2, and 3 respectively. Pins 4 and 5 are connected to pins 4 and 5. Pins 6 through 10 are connected to DIGIT2, DIGIT3, LED2WIRE, DIGIT4, and LED3WIRE respectively.
- H5:** A (pin 1), B (pin 2), SWIO_XIAO (pin 3), RST_XIAO (pin 4), and SWCLK_XIAO (pin 5) are connected to pins 1, 2, 3, 4, and 5 respectively. Pins 6 through 10 are connected to C, D, E, F, and BUZZERWIRE respectively.
- H6:** DP (pin 1), LED1WIRE (pin 2), LDRWIRE (pin 3), SW5WIRE (pin 4), and SW4WIRE (pin 5) are connected to pins 1, 2, 3, 4, and 5 respectively. Pins 6 through 10 are connected to SW2WIRE, G, SW3WIRE, and RSWITCH respectively.

Each diagram includes a red box around the header pins and a red circle around the pin numbers. The text "PM254-2-05-S-8.5" is printed below each diagram.

Signal Buffers

The diagram illustrates two signal buffers, U5.1 and U5.2, both using MCP6002DRG op-amp chips.

U5.1 (Top Buffer):

- Inputs:** LDRWIREBUF (pin 2, inverting input) and LDRWIRE (pin 3, non-inverting input).
- Output:** LDRWIREBUF (pin 1).
- Power:** VCC (pin 8) and GND (pin 4).

U5.2 (Bottom Buffer):

- Inputs:** XIAO3V3BUF (pin 6, inverting input) and XIAO3V3 (pin 5, non-inverting input).
- Output:** XIAO3V3BUF (pin 7).
- Power:** VCC (pin 8) and GND (pin 4).

SWD Header

The diagram illustrates the SWD Header connector. It features a 5-pin header labeled H1 with pins numbered 1 through 4. The connections are as follows:

- Pin 1: VCC
- Pin 2: SWCLK
- Pin 3: SWD
- Pin 4: GND

The component is identified as PZ254R-11-04P.

Schematic	Telemetry Schematic			Create at	2026-09-03
				Update at	2026-09-04
Board	uFerris Megalops Telemetry			Page	P1
Drawn		uFerris			
Reviewed					
		Version	Size	Page 1 Total 1	
		Rev A.0.0	A3	The Embedded Rustacean	